Xiaomin Lin

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Education Background

University of Maryland- College Park (UMD) **Sept/2015-May 2023**
Master of Science, Doctor of Philosophy, Electrical and Computer Engineering GPA 3.51/4.0

University of Dayton (UD) **July/2013-May/2015**
Bachelor of Science, Electrical Engineering GPA 3.75/4.0

Nanjing University of Science and Technology **Sept/2011-May/2015**
Bachelor of Science, Electrical Engineering

Research Interests

Exploring more possibility of conducting research on multi-discipline subjects related to Computer Vision, Natural Language Processing and Robotics.

Research Experience

Research Assistant (Ph.D.) **Sept 2018- Present**

Perception and Robotics Group, UMD advised by Yiannis Aloimonos

- **Unsupervised Learning for Video Action Segmentation.**
  - Developed Deep Recurrent Q-Learning (DRQN) for video action segmentation
  - Introduced the entropy as the reward for the system
  - Built an unsupervised network to recognize the objects
- **Following Instructions by Imagining and Reaching Visual Goals(Preprint)**
  - Organized the simulated environment in Mujoco.
  - Mapped the instructions to the imaging goals using Variational Auto Encoder (VAE)
  - Trained the robot with a low-level policy by directly comparing the encoded goal state with the current state
- **Machine Common Sense for the rationale of numbers (ongoing)**
  - Created a simulated environment for generating random number of known objects.
  - Generated MNIST dataset for 10 types of objects with random number from 0-10
  - Utilized Region Propose Network (RPN) for recognizing the number of objects
  - Worked on Generative Adversarial Network (GAN) for recognizing objects
- **Rearrangement and scene understanding for Embodied AI (ongoing)**
  - Formed a simulated environment with UR-5 robot in Unity and Mujoco
  - Established the Deep Q Learning network with the environment
  - Pioneered the image-torque as the reward
- **Oyster recognition and counting**
  - Created dataset for oyster recognition and used MASK-R-CNN for oyster recognition
  - ***Reward**: Image Recognition Student Challenge 3rd Place from Northrop Grumman*

Research Assistant (Master) **Jan/2016- May 2018**
Autonomous System Lab, UMD

- **Localization of Unmanned Aerial Vehicle with GPS-denied environment.**
  - Structured controllable autonomous unmanned aerial vehicle with on-board Linux computer and mono-camera

- Installed the connections between the vehicle and the transmitter, Raspberry Pi and the vehicle, the drone and the Ground Central Station (Mission Planner)
- Composed a key-points (SURF and SIFT) matching algorithm for basic localization.
- Simulated the algorithm of Terrain Referenced UAV Localization and Navigation
- Collected Dataset and established the Deep Neural Network (DNN) using Tensorflow framework to solve the problem of mono-camera based localization of unmanned aerial vehicle with denied GPS

- **Real-time Personal Following.**
  - Established the platform of a large (150lb+ payload) front wheel steering unmanned ground vehicle.
  - Attained Nodes in ROS for both publishing and subscribing information.
  - Utilized Keras (Python) to implement DNN system for posed-based people-recognition.
  - Developed hardware for electro-mechanical braking and steering and wrote low-level controllers for feedback steering control
  - Made circuitry for power, motor control, and wireless communication
- **Complete Path Coverage Algorithms for Grass Mowing Robot**
  - Simulated the algorithm for path planning algorithms such as A*, Rapidly-Exploring Random Trees, Probabilistic Roadmap Planners
  - Programed in Matlab to simulate the grass mowing problem and implemented the wave-front coverage path planning algorithm into a small unmanned ground vehicle

Undergraduate Research Assistant **Jan/2015-May/2015**

Motoman Robotics Laboratory, UD

- Developed an algorithm that mimics human handwriting on Yaskawa's robot
  - Simulated the algorithm for path planning algorithms such as A*, RRT, and PRM
  - Interpreted the handwriting image captured from camera through Matlab
  - Captured the human handwriting motion from the image
  - Programmed on Yaskawa's robot with Matlab to translate human handwriting image into the corresponding industrial motion

Professional Experience

Bosch Automotive Products Co. Ltd, Robot System Developer Intern **June/2016-Aug/2016**

- Designed a gift drawing showcase in World Manufacturing Conference within PA-ATMO-SGS31 group using Fanuc (APAS) robot
- Set up the Manufacturing Execution System (MES) showcase in World Manufacturing Conference (MFC) with PA-ATMO-SGS21 group
- Collaborated with PA-ATMO-SGS group secretary to make up plans for the MFC and preparation
- Configured MES system for the use for the Chinese customers

Skills

Language and Packages: Python (OpenCV, Tensorflow, Scikit-learn, Keras, drone-kit), Matlab
Platforms: Unix, Linux, Raspbian, Windows
Software: Inventor, Mission Planner, Expert Microsoft Office (Word, Excel, PowerPoint)
Laboratory: Entry level mechanical design and 3D printing, circuit board design and implementation

Relevant Courses

Courses: Computational Linguistics (NLP), Image Understanding, Pattern Recognition, Convex Optimization, Optimal Control, and Aerial Robotics